

Global Supply Chain Risk & Logistics Performance Analysis (SQL + Power BI)

1. Project Overview

This project analyzes global shipment data to understand logistics performance, disruption patterns, and risk factors affecting supply chains.

SQL was used for data exploration and Power BI was used for visualization. The dashboard provides insights into delivery lead times, transport performance, and operational disruptions.

The goal is to help logistics teams identify bottlenecks, assess risk exposure, and improve delivery reliability.

2. Business Problem

Global supply chains face several challenges including weather disruptions, geopolitical risks, transport inefficiencies, and delivery delays. Organizations require data-driven insights to answer key questions:

- Which transport modes are most efficient?
- What factors contribute to shipment disruptions?
- How does geopolitical risk impact delivery times?
- Which products take the longest to deliver?

This analysis helps improve logistics decision-making and risk mitigation strategies.

3. Dataset

Dataset: Global Supply Chain Risk Dataset

Total Records: 5000 shipments

Key Columns:

Column	Description
Shipment_ID	Unique shipment identifier
Date	Shipment date
Origin_Port	Origin location
Destination_Port	Destination location
Transport_Mode	Air, Sea, Rail, Road
Product_Category	Product type

Distance_km	Shipment distance
Weight_MT	Shipment weight
Fuel_Price_Index	Fuel cost indicator
Geopolitical_Risk_Score	Risk score by region
Weather_Condition	Weather condition
Carrier_Reliability_Score	Logistics provider reliability
Lead_Time_Days	Delivery time
Disruption_Occurred	Shipment disruption indicator

4. Tools Used

SQL (MySQL) – Data exploration and analysis

Power BI – Data visualization and dashboard creation

Power Query – Data cleaning and transformation

DAX – KPI calculations

5. Data Preparation

Steps performed before analysis:

1. Imported dataset into MySQL database
2. Created a shipments table

```
CREATE TABLE shipments (
  Shipment_ID VARCHAR(20),
  Date DATE,
  Origin_Port VARCHAR(100),
  Destination_Port VARCHAR(100),
  Transport_Mode VARCHAR(50),
  Product_Category VARCHAR(100),
  Distance_km FLOAT,
  Weight_MT FLOAT,
  Fuel_Price_Index FLOAT,
  Geopolitical_Risk_Score FLOAT,
  Weather_Condition VARCHAR(50),
  Carrier_Reliability_Score FLOAT,
  Lead_Time_Days FLOAT,
  Disruption_Occurred INT
);
```

3. Checked for null values, duplicate records, and data type consistency
4. Loaded cleaned dataset into Power BI
5. Performed additional transformations using Power Query

```
-- Disruption Rate
SELECT
COUNT(*) AS total_shipments,
SUM(Disruption_Occurred) AS disrupted,
SUM(Disruption_Occurred)/COUNT(*)*100 AS disruption_rate
FROM shipments;
```

```
-- Transport Mode Analysis
SELECT
Transport_Mode,
COUNT(*) AS shipments,
AVG(Lead_Time_Days) AS avg_delivery_time,
SUM(Disruption_Occurred)/COUNT(*)*100 AS disruption_rate
FROM shipments
```

```
-- Weather Impact Analysis
SELECT
Transport_Mode,
COUNT(*) AS shipments,
AVG(Lead_Time_Days) AS avg_delivery_time,
SUM(Disruption_Occurred) / COUNT(*) * 100 AS disruption_rate
FROM shipments
GROUP BY Transport_Mode;
```

```
-- Geopolitical Risk Analysis
SELECT
CASE
WHEN Geopolitical_Risk_Score < 3 THEN 'Low Risk'
WHEN Geopolitical_Risk_Score < 6 THEN 'Medium Risk'
ELSE 'High Risk'
END AS risk_level,
COUNT(*) AS shipments,
AVG(Lead_Time_Days) AS avg_lead_time,
SUM(Disruption_Occurred)/COUNT(*)*100 AS disruption_rate
FROM shipments
GROUP BY risk_level;
```

```
-- Carrier Performance
SELECT
Carrier_Reliability_Score,
AVG(Lead_Time_Days) AS avg_lead_time,
SUM(Disruption_Occurred)/COUNT(*)*100 AS disruption_rate
FROM shipments
GROUP BY Carrier_Reliability_Score
ORDER BY Carrier_Reliability_Score DESC;
```

```
-- Route Analysis
SELECT
Origin_Port,
Destination_Port,
COUNT(*) AS shipments,
SUM(Disruption_Occurred) AS disruptions
FROM shipments
GROUP BY Origin_Port, Destination_Port
ORDER BY disruptions DESC
LIMIT 10;
```

```
-- Lead Time Analysis
SELECT
Product_Category,
AVG(Lead_Time_Days) AS avg_delivery_time
FROM shipments
GROUP BY Product_Category
ORDER BY avg_delivery_time DESC;
```

```
-- Distance Impact
SELECT
CASE
WHEN Distance_km < 2000 THEN 'Short'
WHEN Distance_km < 8000 THEN 'Medium'
ELSE 'Long'
END AS distance_category,
AVG(Lead_Time_Days) AS avg_lead_time,
SUM(Disruption_Occurred)/COUNT(*)*100 AS disruption_rate
FROM shipments
GROUP BY distance_category;
```

6. Key KPIs

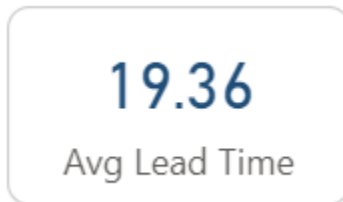
- Total Shipments – Total number of shipments processed



- Total Shipment Disruptions – Number of disrupted shipments



- Average Delivery Lead Time – Average shipment delivery duration



- Shipment Disruption Rate – Percentage of disrupted shipments



- Average Carrier Reliability Score – Logistics provider reliability

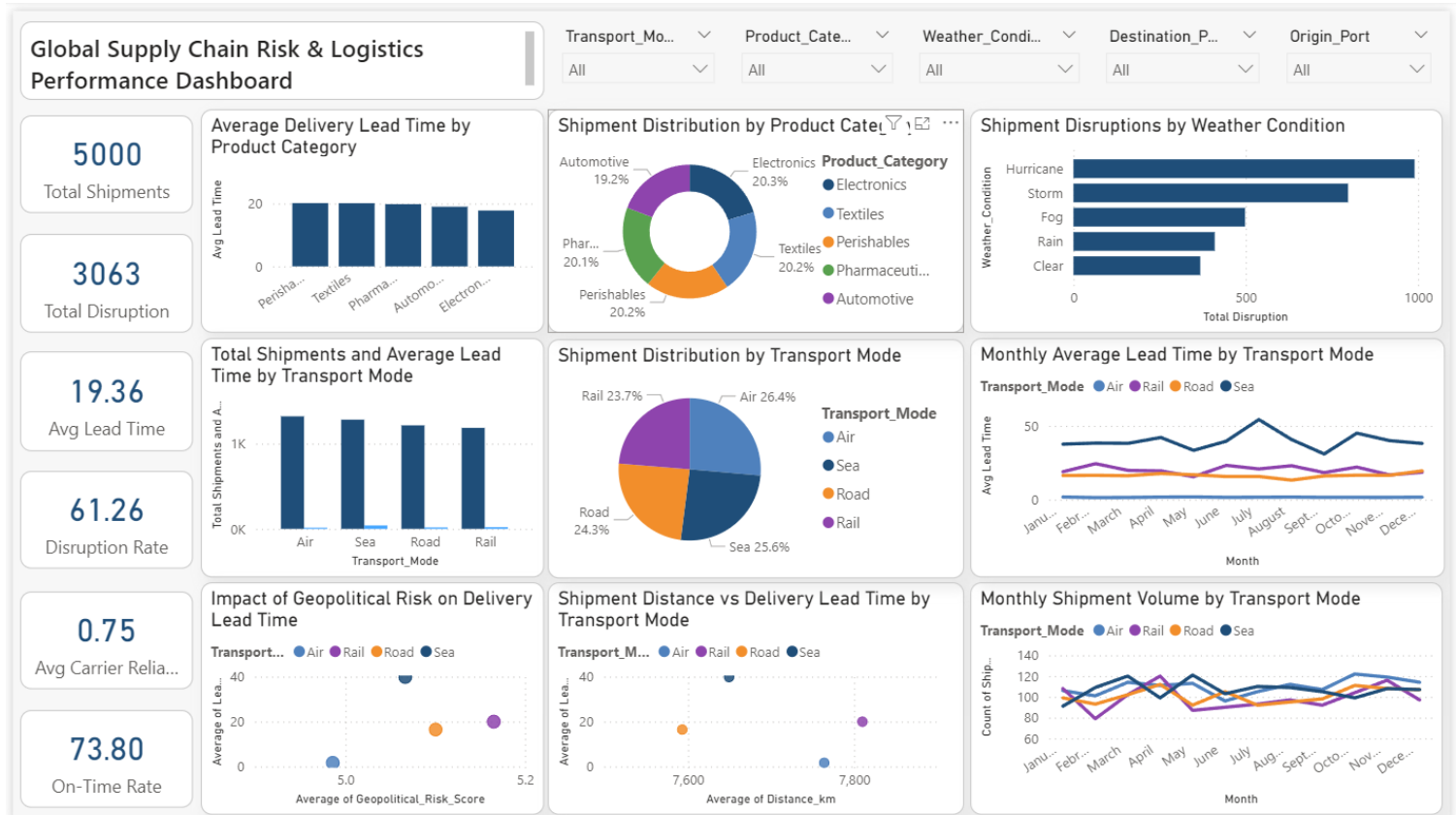


- On-Time Delivery Rate – Percentage of shipments delivered within threshold



7. Dashboard Visualizations

The Power BI dashboard includes the following visualizations:



- Average Delivery Lead Time by Product Category
- Shipment Distribution by Product Category
- Shipment Disruptions by Weather Condition
- Total Shipments and Average Lead Time by Transport Mode
- Shipment Distribution by Transport Mode
- Monthly Average Lead Time by Transport Mode
- Impact of Geopolitical Risk on Delivery Lead Time
- Shipment Distance vs Delivery Lead Time
- Monthly Shipment Volume by Transport Mode

8. Key Insights

1. Sea transport shows the highest delivery lead times, making it less suitable for time-sensitive shipments.
2. Extreme weather conditions such as hurricanes and storms cause the highest number of shipment disruptions.
3. Higher geopolitical risk scores correlate with longer delivery lead times.
4. Air transport manages long-distance shipments while maintaining the fastest delivery times.
5. Shipment volume is relatively balanced across transport modes.

9. Business Recommendations

- Increase the use of air transport for high-priority shipments.
- Improve contingency planning for weather-related disruptions.
- Monitor shipments in high-risk geopolitical regions.
- Evaluate logistics partners to improve carrier reliability.

10. Dashboard Features

- Interactive slicers for Transport Mode, Product Category, Weather Condition, Origin Port, and Destination Port
- Dynamic filtering across all visuals
- KPI tracking
- Trend analysis

11. Project Folder Structure

```
supply-chain-risk-analysis
|
├── dataset
|   └── global_supply_chain_risk_2026.csv
|
├── sql
|   └── analysis_queries.sql
|
├── powerbi
|   └── supply_chain_dashboard.pbix
|
└── README.md
```

Rushikesh Vilas Patil Data Analyst — Python | SQL | Power BI

 Email: rushiart77492@gmail.com

 GitHub: github.com/RusshikeshPatil

 LinkedIn: linkedin.com/in/russhikesh-patil